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CS 621 DATA ANALYTICS

Data Engineering

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DATA ENGINEERING

- Under different names like:
  - Data Munging, Checking, Filtering, Validation, Cleaning, Manipulation
- Check for data errors.
  - Remove erroneous data.
  - Fix erroneous data.
- Make data ready for specific tasks.
  - May need to have a complete set of predefined rules.

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COMMON DATA ERRORS

- Data type errors for
  - Numerical values
  - String values
- Out of range errors
  - check for the min and max bounds
- Data inconsistency
  - Date: 02-11-20, 02/11/2020, Feb 2, 2020
  - Boolean values: True, T, 1

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MISSING/EMPTY VALUES

- Check for null/empty/missing values.
    - What to do with them:
        - Remove the whole record.
        - Replace the null value with something pre-defined.
- NaN (Not a Number)
    - a floating point number
    - cannot be used for direct comparison.
    - use math.nan, numpy.nan to assign NaN.
    - use math.isnan(), pandas.isna(), numpy.isnan(), series.isnull(), dataframe.isnull() to detect NaN.

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DUPLICATES

- How to check duplicates?
  - Duplicated Columns
  - Duplicated Rows
  - Use for loops to compare elements.
  - Both Pandas Series and Dataframe have .duplicated() method.
- What to do with duplcates?
  - Use .drop_duplicates() method in both Series and DataFrame.
  - Keep one copy of the duplicates (can be first or last).
  - Remove all the duplicates.

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DATA INTEGRATION

- Integrate data from
  - different sources
  - different data structures
  - different data formats
- Merge data
  - in same data structure
  - in different data structures

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DATA TRANSFORMATION

- Different formats
  - JSON <-> XML
- Different units
  - temp: °C <-> °F
- Different scales
  - Months -> Year



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WE FLY